# Kieran Fraser, PhD

Staff Machine Learning Engineer. Production ML, adversarial AI security and fraud detection. Dublin, Ireland.



## Summary

Staff Machine Learning Engineer with a PhD in Computer Science and eight-plus years spanning production machine learning, adversarial AI security and applied research. Currently build and operate fraud-detection models at PayPal; previously a Research Engineer on IBM's Security and Privacy of AI team, contributing to open-source adversarial-robustness and LLM red-teaming tools (HEART, ART, ARES) and co-authoring research on LLM guardrails, jailbreak detection and synthetic data generation. Experienced translating research into deployed systems and customer-facing outcomes — leading an Enterprise Ireland-funded commercialisation project end to end, mentoring engineers, and shaping technical direction through a company patent programme.

## Core skills

**Machine learning & AI** — machine learning, deep learning, large language models (LLMs), generative AI, adversarial robustness, AI red-teaming & guardrails, synthetic data generation, agentic AI, fraud detection, MLOps

**Engineering** — Python, Java, C++, JavaScript, SQL & NoSQL, Docker, Node.js, React, HTML5/CSS/SCSS

**Cloud & infrastructure** — AWS (SageMaker, Lambda, S3, DynamoDB, API Gateway, ECR, Batch, EC2, IAM, Cognito, Athena, Glue, Amplify), Google Cloud Platform, Microsoft Azure

**Leadership & delivery** — technical mentorship, cross-functional collaboration, patent development, project management, customer & trial-partner engagement,

academic research & publication

## Employment history

**Staff Machine Learning Engineer, PayPal** NOV 2026 – PRESENT · DUBLIN

- Build and maintain production machine learning models for onboarding and early-lifecycle fraud detection, balancing fraud prevention against customer experience at scale.
- Contributed to a redesign of the team's model portfolio and evaluation approach.
- Filed two patent applications and founded a cross-team patent group to help colleagues navigate the IP process.
- Mentor two engineering interns.

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**Research Engineer, IBM Research** AUG 2022 – NOV 2026 · DUBLIN

- Contributed to HEART, ART and ARES — IBM's open-source toolkits for adversarial-robustness evaluation and automated LLM red-teaming — including 104 commits to HEART per its public contributor record.
- Co-authored peer-reviewed and arXiv research on LLM guardrails, jailbreak detection and red-teaming methodology (Granite Guardian, MoJE, MAD-MAX, Attack Atlas), plus synthetic data generation and agentic-AI workload management.
- Named inventor on four published patent applications spanning federated learning, retrieval-augmented generation security, visual-content anonymisation and model-capability analysis.
- Represented IBM at IEEE ERAS 2025 (Worcester Polytechnic Institute) and AI4Media's demonstration sessions at the University of Florence.

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**Research Assistant, ADAPT Centre for Digital Content Technology** SEP 2020 – AUG 2022 · DUBLIN

- Led an Enterprise Ireland-funded Feasibility Study and Commercialisation project (EmPushy) in empathetic push-notifications, recruiting and managing a small team.
- Built and deployed an MVP; analysed 3M+ real-world notifications and presented findings directly to partner CTOs and R&D leads.

- Managed AWS infrastructure, GDPR compliance and IP documentation for the project.

### Earlier experience

2013 – 2016

Android Engineer at Lumafit (2016) — migrated a local Android database to a real-time cloud backend. Executive Officer at the Revenue Commissioners (2015) — rebuilt internal web applications with the Spring Framework. Data Analyst at the Aon Centre for Innovation & Analytics (2014) — built a BI dashboard (SQL, Tableau) presented to ACIA's CEO, CBO and CCO. Post-processing Engineer at Septentrio, Belgium (2013) — contributed to in-house testing and CI (Jenkins) for satellite-navigation software.

## Education

### PhD Computer Science, Trinity College Dublin

SEP 2016 – MAR 2021 · DUBLIN

Thesis: *ETHOS Push* — EmpaTHetic Orchestration of Scrutable Push. A framework for transparent, context-aware, empathetic push-notification delivery.

### MAI Computer Engineering, Trinity College Dublin

SEP 2015 – MAY 2016 · DUBLIN

Graduated with Distinction.

### BAI Computer Engineering & BA Mathematics, Trinity College Dublin

SEP 2011 – MAY 2015 · DUBLIN

Graduated with Second Class Honours, First Division (2.1).

## Selected publications

19 publications in total. Full list on Google Scholar or the Work page.

### Granite Guardian: Comprehensive LLM Safeguarding

I. PADHI, M. NAGIREDDY, G. CORNACCHIA, ET AL. · NAACL 2025

### Attack Atlas: A Practitioner's Perspective on Challenges and Pitfalls in Red Teaming GenAI

A. RAWAT, S. SCHOEPF, G. ZIZZO, G. CORNACCHIA, K. FRASER, ET AL. · 2024

MoJE: Mixture of Jailbreak Experts, Naive Tabular Classifiers as Guard for Prompt Attacks

G. CORNACCHIA, G. ZIZZO, K. FRASER, ET AL. · AAAI/ACM AIES 2024

*ETHOS Push: An Approach toward Empathetic Orchestration of Scrutable Push* — PhD thesis, Trinity College Dublin

2021

A Context-Aware, Info-Bead and Fuzzy Inference Approach to Notification Management

K. FRASER, B. YOUSUF, O. CONLAN · UEMCON 2016 · BEST PAPER AWARD

## Patents

Named inventor on four published IBM applications and two further applications filed at PayPal (not yet public). Full record on Google Patents.

Securing retrieval-augmented generation IBM

Model emergent capability analysis IBM

## Hobbies

CrossFit, golf, tag rugby, art, photography, VR and photogrammetry — more on the Elsewhere page.

## Links

LinkedIn

GitHub

Google Scholar